

Dr. Auqib Hamid Lone

BLOCKCHAIN RESEARCHER · APPLIED CRYPTOGRAPHY RESEARCHER

Dalwath, Qazigund, Anantnag, Jammu and Kashmir, India, 192221

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“Security is the balanced mix of feeling and reality.”

Education

National Institute of Technology

Hazratbal, Srinagar, India

PHD. IN COMPUTER SCIENCE AND ENGINEERING (BLOCKCHAIN TECHNOLOGY)

2017 - 2021

- In my Ph.D., I explored Blockchain Technology Applications beyond Cryptocurrencies. In particular, I worked towards harnessing Blockchain technology to eliminate the problem of trust and security in various applications.

Jamia Hamdard

New Delhi, India

M.TECH. IN COMPUTER SCIENCE AND ENGINEERING (INFORMATION SECURITY & CYBER FORENSICS)

2014 - 2016

- I secured 1st Rank in my Master's degree program.

BGSBU(Baba Ghulam Shah Badshah University)

Rajouri, J&K

B.TECH. IN INFORMATION TECHNOLOGY

2010 - 2014

- I was awarded a Gold medal during the degree program for exceptional performance in my batch.

Experience

GCET Kashmir

Ganderbal, Kashmir, India

ASSISTANT PROFESSOR (COMPUTER SCIENCE & ENGINEERING)

March 2024 – Present

- Teaching core and advanced undergraduate courses to B.Tech CSE students.

Parity Financial

London, UK

RESEARCH SCIENTIST (BLOCKCHAIN AND CRYPTOGRAPHY)

April 2021 – Feb 2023

- Worked on improving underlying mechanisms in cryptography, blockchain, and security systems.
- Led integration of new blockchain networks into custody infrastructure, gaining hands-on expertise in transaction construction, signing workflows, and key management.
- Core contributor to the initial concept of Rayls—an EVM-compatible privacy layer using zero-knowledge proofs (Rayls Enygma) enabling confidential transactions with selective disclosure.
- Conducted research and developed PoCs to support integration of new blockchain platforms and enhance existing systems.

BloomTech USA and Premier Engineering Institutes in India

SUBJECT MATTER EXPERT (PART-TIME)

April 2019 – 2022

- Delivered training sessions on blockchain technology and developed a structured curriculum for engineering students.
- Served as SME for Web3 courses, specializing in Ethereum token standards (ERC20, ERC721).

Honors & Awards

2018	NTA NET-JRF , in Computer Applications	99.9%ile
2018	GATE , in CSIT	
2016	1 st Rank , Master of Technology	New Delhi, India
2014	Gold Medalist , Bachelor of Technology	Rajouri, J&K
2013	1 st Position , Geeks Fest MMXII	BGSBU, Rajouri,
2013	2 nd Prize , Quiz Competition organized by University Polytechnic	Rajouri
2013	3 rd Prize , Quiz Competition organized by Dept. of Information Technology	Rajouri

Publications

- 2020 **Demystifying Cryptography behind Blockchains and a Vision for Post-Quantum Blockchains** , (2020 IEEE International Conference for Innovation in Technology (INOCON))
- 2021 **Applicability of Blockchain smart contracts in securing Internet and IoT: A systematic literature review**, (Computer Science Review Volume 39, Feb 2021)
- 2020 **Forgery Protection of Academic Certificates through Integrity Preservation at Scale using Ethereum Smart Contract**, (Scalable Computing, Dec. 2020)
- 2020 **Applicability of mobile contact tracing in fighting pandemic (COVID-19): Issues, challenges and solutions**, (Computer Science Review Volume 38, November 2020)
- 2020 **Blockchain Driven Access Control Mechanisms, Models and Frameworks: A Systematic Literature Review**, (IACR Cryptol. ePrint Arch. 2020,1379)
- 2020 **Evaluating Quality-of-Service in Blockchains Using Modelling and Simulation Tools** , (International Journal of Computing and Digital Systems 10, 2-8)
- 2020 **Reputation Driven Dynamic Access Control Framework for IoT atop PoA Ethereum Blockchain**, (IACR Cryptol. ePrint Arch. 2020, 566)
- 2019 **Forensic-chain: Blockchain-based digital forensics chain of custody with PoC in Hyperledger Composer** **AH Lone, RN Mir**, (Digital Investigation 28, 44-55)
- 2018 **Consensus protocols as a model of trust in blockchains** **AH Lone, RN Mir**, International Journal of Blockchains and Cryptocurrencies 1 (1), 7-21
- 2017 **Investigating and Analyzing Bitcoin Blockchain Protocol using Wireshark** **AH Lone, RN Mir**, International Journal of Computer Network and Information Security 10 (7), 36
- 2017 **Comparative Analysis of SLA-LFSR with Traditional Pseudo Random Number Generators** , International Journal of Computational Intelligence Research ISSN 0973-1873 Volume 13, Number 6, pp. 1461-1470
- 2017 **A Novel Scheme for Image Authentication and Secret Data Sharing**, International Journal of Computer Networks and Information Security
- 2017 **Acoustic lightweight pseudo-random number generator based on cryptographically secure LFSR MAS** **AL-khatib, AH Lone**, International Journal of Computer Network and Information Security 11 (2), 38
- 2016 **Common Attacks on RSA and its Variants with Possible Countermeasures**, International Journal of Emerging Research in Management & Technology ISSN: 2278-9359 (Volume-5, Issue-5)
- 2016 **Generalized RSA Using $2k$ Prime Numbers with Secure Key Generation**, Security and communication networks WILEY
- 2015 **Implementation of Forensic Analysis Procedures for WhatsApp and Viber Android Applications**, International Journal of Computer Applications Volume 128/Number 12 (ISBN: 973-93-80889-79-2)
- 2015 **On the Applicability of Genetic Algorithms for Cryptanalysis: A Review** , International Journal of Computer Applications Volume 130 – No.9 (0975 – 8887)
- 2015 **A Novel Unpredictable Temporal-based Pseudorandom Number Generator**, International Journal of Computer Applications (0975-8887) Volume 117-No.13

Workshops & Conferences

Workshop on Network Security

STUDENT COORDINATOR

CoET, BGSSBU, Rajouri

Sept. 2012

National Level Workshop Robospecies

STUDENT PARTICIPATION

CoET, BGSSBU, Rajouri

Oct. 2012

HACK ATTACK Training programme on Information Security & Ethical Hacking

STUDENT PARTICIPATION

*Rational Tabs Private Ltd Srinagar
Kashmir*

Jan. 2013

National Ethical Hacking Workshop under Tech Bharat consulting

STUDENT PARTICIPATION

IMS Jammu

Mar. 2013

Workshop on RFID based Pervasive Computing

STUDENT COORDINATOR

CoET, BSBU, Rajouri

April. 2013

Training's & Certifications

NPTEL15CS031801942

INTRODUCTION TO INFORMATION SECURITY-1

NPTEL, IIT Madras, India

2016

National Institute of Electronics & Information Technology

PL/SQL(OCA) PROGRAMMING

Srinagar, India

2013

Coursera-University of Buffalo

BLOCKCHAIN BASICS

[https://coursera.org/share/
9e03d9aa3991c90d7f248841f1365e1a](https://coursera.org/share/9e03d9aa3991c90d7f248841f1365e1a)

September, 2020

Coursera-University of Buffalo

BLOCKCHAIN PLATFORMS

[https://coursera.org/share/
1986538372b8c524c74cb1d97b38c79d](https://coursera.org/share/1986538372b8c524c74cb1d97b38c79d)

September, 2020

Coursera-University of Buffalo

SMART CONTRACTS

[https://coursera.org/share/
095df021d1d273b0072e704d5df6859f](https://coursera.org/share/095df021d1d273b0072e704d5df6859f)

September, 2020

Coursera-University of Buffalo

DECENTRALIZED APPLICATIONS (DAPPS)

[https://coursera.org/share/
8c1628ebd6e1a10524d0041b119948db](https://coursera.org/share/8c1628ebd6e1a10524d0041b119948db)

September, 2020